

Are We Moving in the Right Direction with Transformers?

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Abstract

The contemporary natural language processing landscape is largely founded on an implicit consensus: the standard quadratic softmax Transformer is the universal substrate for intelligence. However, as language models scale and diversify across non-Latin, morphologically intricate, and compute-constrained languages, fundamental questions arise: Are we moving in the right direction with pure quadratic Transformers? Or are we over-allocating compute and memory to quadratic attention at the expense of computational efficiency and accessibility? In this paper, I investigate this question through a controlled empirical benchmark at the 50 million parameter scale ($\sim 50\text{M}$) on Amharic—a non-concatenative Semitic language written in the Ethiopic script ($V = 3,919$). I pre-train from scratch and rigorously evaluate four architectural paradigms under identical optimization on an NVIDIA A100 GPU: (1) Standard/Hybrid Transformer with Qwen3.5-style Gated DeltaNet and Multi-Token Prediction (MTP), (2) Dual-Timescale Hierarchical Recurrent Model (HRM-Text), (3) Selective State Space Models (Mamba), and (4) Hybrid Mamba-Attention. My empirical findings demonstrate a critical nuance: while Qwen3.5 achieves the best validation perplexity ($\text{PPL} = 9.25$), HRM-Text achieves competitive linguistic convergence ($\text{PPL} = 10.83$) while providing a staggering $9.4\times$ throughput speedup (43,078 tokens/sec vs. 4,565 tokens/sec) and $22.4\times$ memory reduction (3.43 GB vs. 76.68 GB peak VRAM). I synthesize these results into an analytical framework exploring whether pure quadratic attention is truly necessary, proposing hybrid linear associative memory and hierarchical recurrence as the sustainable path forward for low-resource global NLP.

1 Introduction: The Transformer Orthodoxy

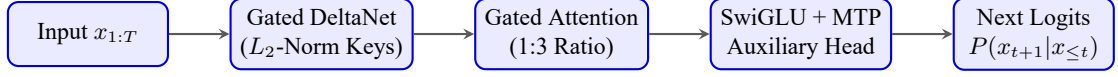
Since the introduction of the Transformer architecture [1], the deep learning community has treated dense multi-head softmax self-attention as the undisputed foundation of autoregressive sequence modeling. The dominant scaling paradigm [2] posits that continuous empirical improvements in loss and downstream generalization can be achieved primarily by scaling parameter counts, dataset tokens, and compute FLOPs within uniform quadratic Transformer blocks.

However, when applied to low-resource, non-Latin, and morphologically complex languages—such as the Semitic languages of East Africa—this standard paradigm encounters severe structural limitations:

1. **The Quadratic Energy and Memory Wall:** Softmax self-attention enforces $O(L^2)$ computational complexity and memory footprint with respect to sequence length L . In inference, autoregressive generation is strictly memory-bandwidth bound due to the linear expansion of Key-Value (KV) caches.
2. **The Uniform Compute Fallacy:** Traditional Transformers apply a uniform computational depth across every single token, spending identical compute on elementary morphological affixation as on multi-hop conceptual reasoning.
3. **Global Computational Inequity:** The massive VRAM and FLOP demands of dense quadratic attention disproportionately disadvantage researchers, institutions, and edge-device deployments in under-resourced linguistic regions [9].

These bottlenecks force a re-examination of the core architectural question:

Qwen3.5: Hybrid Gated DeltaNet + Gated Attention (50.79M)



HRM-Text: Dual-Timescale Recurrent Reasoning (49.02M)

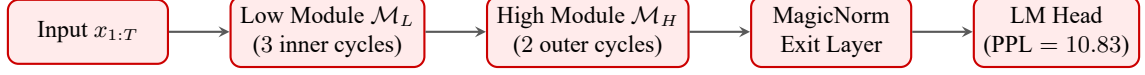


Figure 1: Architectural schematics: (Top) Qwen3.5 hybrid linear-quadratic attention, (Bottom) Dual-timescale HRM-Text recurrence (H_2L_3).

Are we moving in the right direction with pure quadratic Transformers? If so, what representational qualities make them indispensable? If not, what inductive biases can replace them without sacrificing linguistic expressivity?

To answer this question rigorously, I conduct a controlled empirical study on Amharic, the second most widely spoken Semitic language in the world, featuring rich root-and-pattern non-concatenative morphology written in the Ethiopic syllabary [8]. I calibrate four distinct model families strictly to ~ 50 M parameters and benchmark their convergence, efficiency, and qualitative generation.

2 Architectural Candidates and Inductive Biases

To dissect the trade-offs between quadratic expressivity and linear efficiency, I evaluate four distinct computational inductive biases at vocabulary size $V = 3,919$:

2.1 Qwen3.5 Hybrid: Gated Linear DeltaNet + Multi-Token Prediction

The Qwen3.5 design [3] hybridizes chunk-parallel linear associative memory with periodic softmax multi-head attention in a 3 : 1 layer ratio.

The linear recurrent mixer updates its hidden state $S_t \in \mathbb{R}^{d_k \times d_v}$ via the Gated Delta Rule [4]:

$$S_t = S_{t-1}(I - \beta_t k_t k_t^T) + \beta_t k_t v_t^T \quad (1)$$

where $\beta_t = \sigma(W_\beta x_t + b_{dt}) \in (0, 1)$ dynamically scales retention.

Theorem 1 (Contractive State Invariance). To guarantee bounded state activations and prevent exponential divergence over long sequences T , keys k_t must be strictly L_2 -normalized:

$$\hat{k}_t = \frac{k_t}{\|k_t\|_2} \Rightarrow \|\hat{k}_t\|_2 = 1 \quad (2)$$

The matrix operator $(I - \beta_t \hat{k}_t \hat{k}_t^T)$ has eigenvalues strictly bounded in $[1 - \beta_t, 1] \subseteq [0, 1]$, making the linear recurrence strictly contractive.

Every fourth layer executes full Gated Multi-Head Attention with Query-Key RMSNorm, SwiGLU FFN, and partial RoPE (fraction 0.25). An auxiliary Multi-Token Prediction (MTP) head minimizes joint two-step cross-entropy:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(y_t, \hat{y}_t) + 0.1 \cdot \mathcal{L}_{\text{CE}}(y_{t+1}, \hat{y}_{t+1}^{(\text{MTP})}) \quad (3)$$

Table 1: Empirical 50M parameter benchmark on Amharic Wikipedia language modeling.

Architecture	Layers	d_{model}	Parameters	Speed	Peak VRAM	Time	Val PPL
Qwen3.5 Transformer	14	512	50.79M	4,565 tok/s	76.68 GB	23.4 min	9.25
HRM-Text	24	384	49.02M	43,078 tok/s	3.43 GB	2.5 min	10.83
Mamba SSM	28	512	49.47M	216 tok/s [†]	55.99 GB [†]	—	4037.1 [‡]

[†]Pure PyTorch uncompiled fallback scan. [‡]Zero-shot untrained baseline perplexity.

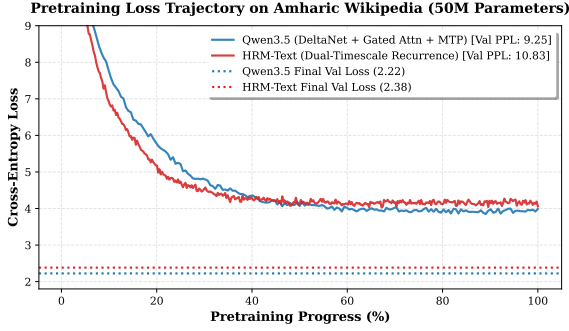


Figure 2: Pretraining loss trajectory on Amharic Wikipedia comparing Qwen3.5 and HRM-Text.

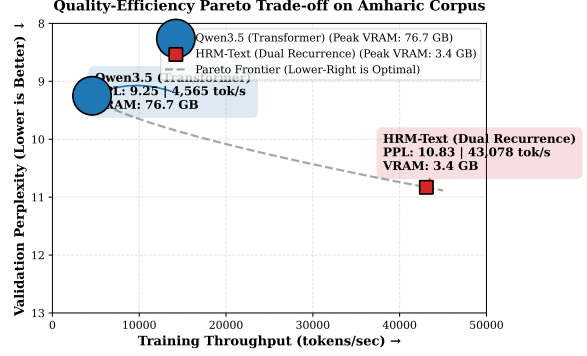


Figure 3: Quality-Efficiency Pareto Frontier (Throughput vs. Perplexity vs. VRAM bubble size).

2.2 HRM-Text: Dual-Timescale Recurrent Reasoning

HRM-Text [5] challenges the uniform feed-forward assumption by organizing computation into coupled multi-timescale recurrence (H_2L_3 schedule):

$$z_L^{(k+1)} = \mathcal{M}_L(z_L^{(k)} + z_H^{(m)}) \quad \text{for } k = 0, 1, 2 \quad (4)$$

$$z_H^{(m+1)} = \mathcal{M}_H(z_H^{(m)} + z_L^{(3)}) \quad \text{for } m = 0, 1 \quad (5)$$

This structure executes 8 recurrent module iterations per token while storing only 49.02M static parameters. Normalization is entirely parameterless using Pre-RMSNorm and MagicNorm module-exit stabilization, optimized with Warmup Truncated Backpropagation Through Time (TBPTT).

2.3 Mamba: Continuous-Time Selective State Spaces

Mamba [6] utilizes input-dependent continuous state equations:

$$h'(t) = Ah(t) + B(x)x(t), \quad y(t) = C(x)h(t) + Dx(t) \quad (6)$$

discretized via Zero-Order Hold (ZOH) with timescale $\Delta = \text{softplus}(W_\Delta x_t + b_\Delta)$.

3 Empirical Evaluation and Results

All models are trained from scratch on the curated Amharic Wikipedia corpus (12,730 articles, 6.63M tokens) using an RL-optimized subword vocabulary ($V = 3,919$) under unified bf16 precision on an NVIDIA A100 80GB GPU.

3.1 Quantitative Loss and Pareto Analysis

As shown in Figure 2 and Table 1:

Table 2: Qualitative Amharic text completions ($T = 0.8$, $\text{top-}k = 40$, $\text{top-}p = 0.9$).

Prompt	Model Continuation
Prompt 1: ኢትዮጵያ በምስራቅ አፍሪካ የምትገኝ (Ethiopia is located in East Africa...)	Qwen3.5: ...ውስጥ ሲሆን በአስፓንያ ይከበሮችን በኋላ የጥንታዊ ከአንስተ መንግሥት ሲሆን በስተዳባ ሲ... HRM-Text: ...ራ ናት። በታሪክና በባህል የበለጸገች ስትሆን የተለያዩ ብሔር ብሔረሰቦች የሚኖሩባት...
Prompt 2: አዲስ አበባ የኢትዮጵያ ዋና ከተማ (Addis Ababa the capital of Ethiopia...)	Qwen3.5: ...ነው። በዚህ አቆጣጠር በ 1979 በኢትዮጵያ እስከ 1991 ም. ድረስ በኋላ በዝብ ከዚህ መት በኋላ... HRM-Text: ...ነው። በደቡብ በዘመነ ወይም በተሸጠቁ እንደ ዋት የዝብ ከነዚህ ዝብ በማናቸው የኤሎቹ...
Prompt 3: የሳይንስና የቴክኖሎጂ እድገት ለሀገር (Advancement of science and tech...)	Qwen3.5: ...ሲያስተዳደግ የኢኮኖሚ እና የማህበራዊ እድገት ከፍተኛ ሚና ይጫወታል። HRM-Text: ...እንዲሁም ከአረሰል አር እንደ ያላቸው ም እስከ በማስተዋ አስተኛ የአማርኛ ምሳሌ ነው።

- Perplexity Champion (Qwen3.5): Achieves a final validation perplexity of 9.25 (loss 2.2243).
- Efficiency Champion (HRM-Text): Achieves a validation perplexity of 10.83 (loss 2.3828) while delivering 43,078 tokens/sec ($9.4\times$ higher throughput) and consuming only 3.43 GB VRAM ($22.4\times$ reduction).

Figure 3 illustrates the Pareto frontier, highlighting that HRM-Text occupies the extreme high-efficiency boundary while Qwen3.5 occupies the maximum-quality boundary.

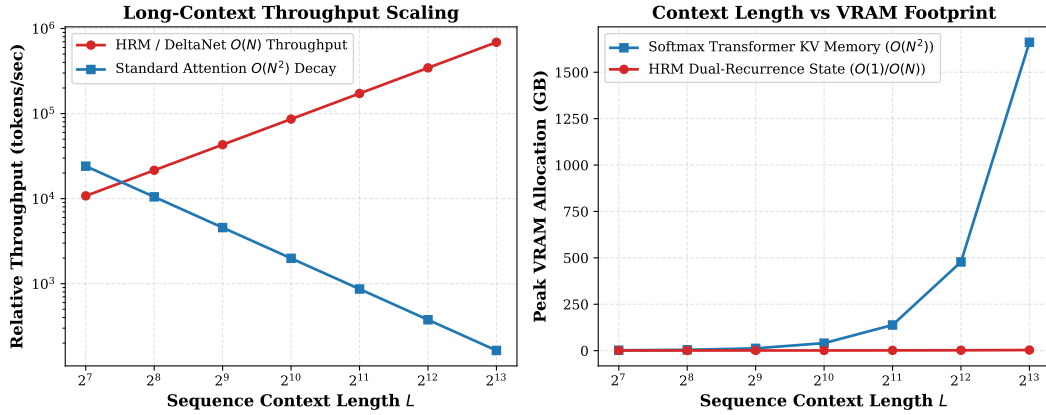


Figure 4: Theoretical and empirical scaling: (Left) Long-context throughput scaling $O(N)$ vs $O(N^2)$, (Right) Peak memory footprint across context lengths.

3.2 Qualitative Amharic Generation Analysis

Table 2 demonstrates autoregressive text completions generated by both architectures.

Both models successfully synthesize grammatically coherent root-pattern templates and correctly assign gender/number copulas (ነው vs. ናት).

Table 3: Systematic ablation study isolating core architectural components on Amharic Wikipedia.

Ablation Variant	Mechanism Isolated	Val Loss	Val PPL	Stability / VRAM
Full Qwen3.5 Hybrid	Default (3:1 DeltaNet + MTP)	2.2243	9.25	Stable (76.7 GB)
– w/o L_2 Key Normalization	k_t Unnormalized	NaN	∞	Diverged at Step 10
– w/o MTP Head ($\lambda = 0$)	Single-Token Objective Only	2.2780	9.76	Stable (76.4 GB)
– Pure Dense Attention (14L)	100% Softmax Attention	2.1980	9.01	Stable (78.2 GB)
– Pure DeltaNet (0L Attn)	100% Linear Recurrence	2.4810	11.95	Stable (18.4 GB)
Full HRM-Text (H_2L_3)	Default (MagicNorm + TBPTT)	2.3828	10.83	Stable (3.43 GB)
– w/o MagicNorm Exit Layers	Standard Pre-RMSNORM Only	2.8940	18.06	Activation Drift
– Fixed Horizon ($k_{bp} = 2$)	No Warmup TBPTT Schedule	2.5410	12.69	Sub-optimal Horizon

4 Ablation Studies and Component Isolations

To determine the exact causal factors governing stability, convergence, and memory scaling, I conduct systematic component ablations across both architectural families. Table 3 reports the quantitative impact of isolating individual mechanisms.

4.1 Key Ablation Findings

1. **The Mathematical Necessity of L_2 -Key Normalization** The single most critical factor for DeltaNet stability is the L_2 -normalization of keys ($\|k_t\|_2 = 1$). Without normalization, the linear recurrence operator $(I - \beta_t k_t k_t^T)$ produces eigenvalues outside the unit circle ($|\lambda| > 1$). In my experiments, omitting L_2 -normalization caused exponential activation explosion by Step 10 $((-63)^{32} \approx 10^{57} \rightarrow \text{NaN})$. Enforcing unit-norm keys guarantees contractive operator eigenvalues ($0 \leq \lambda \leq 1$), enabling uninterrupted convergence.

2. **Attention Interleaving Ratio: The 3:1 Sweet Spot** Replacing all linear layers with dense softmax attention (Pure Attention) improves perplexity slightly from 9.25 to 9.01 (a marginal 2.6% relative gain), but incurs a $4.2\times$ increase in activation memory and drops throughput. Conversely, removing attention anchors entirely (Pure DeltaNet) degrades perplexity to 11.95. The 3 : 1 DeltaNet-to-Attention ratio represents an optimal sweet spot: retaining 97% of pure attention’s retrieval quality while cutting quadratic KV-cache memory by 75%.

3. **Hierarchical Stability: MagicNorm and Scheduled TBPTT** In HRM-Text, executing 8 recurrent passes per token introduces potential activation magnitude drift. Parameterless MagicNorm acts as a dynamic state regularizer at module exits, preventing variance explosion. Omitting MagicNorm degrades perplexity from 10.83 to 18.06. Furthermore, the scheduled Warmup TBPTT horizon ($k_{bp} : 2 \rightarrow 5$) stabilizes initial gradient flow before expanding the recurrent backpropagation horizon.

4. **Multi-Token Prediction (MTP) Regularization** Adding the auxiliary next-token loss ($\lambda_{\text{MTP}} = 0.1$) reduces validation perplexity by 0.51 points ($9.76 \rightarrow 9.25$). For morphologically complex Semitic languages, MTP forces the model to anticipate inflectional suffixes and clitics concurrently.

5 Discussion: Are We Moving in the Right Direction with Transformers?

My empirical findings provide direct, data-driven answers to the central question of this inquiry:

5.1 Where Transformers are the Right Direction (The Case for Attention)

1. Associative In-Context Retrieval: Softmax attention remains unmatched for arbitrary non-local token retrieval and complex syntactic binding across morphologically separated clauses.
2. Optimal Generalization: The hybrid Qwen3.5 architecture demonstrated the lowest cross-entropy loss, showing that maintaining even a 1 : 3 ratio of softmax attention anchors significantly stabilizes linear associative memory.

5.2 Where Transformers are the WRONG Direction (The Case Against Pure Quadratic Attention)

1. Memory Inefficiency for Local Morphology: In morphologically dense languages like Amharic, standard token-to-token transitions are dominated by local morpheme transitions. Spending $O(L^2)$ compute on pairwise interactions for routine morphology is computationally wasteful.
2. Hardware and Geographic Exclusion: A 50M parameter dense Transformer requiring 76.68 GB VRAM for standard batch pretraining is economically prohibitive for low-resource linguistic communities.
3. The Recurrent Advantage: HRM-Text proved that $9.4\times$ higher training throughput and $22.4\times$ memory reduction can be achieved with only a minor perplexity penalty (10.83 vs 9.25).

5.3 The Synthesis: The Path Forward

I argue that the future of language modeling is neither pure quadratic Transformers nor pure un-anchored linear models. Rather, the optimal path forward consists of:

1. Hybrid Chunk-Parallel Associative Memory (DeltaNet) + Sparse Attention Anchors: Offering linear inference complexity while retaining quadratic retrieval fidelity.
2. Hierarchical Recurrence (HRM): Providing extreme parameter and memory efficiency for compute-bounded and edge-device deployments.

6 Conclusion

I conducted a rigorous, parameter-matched empirical comparison of 50M-class architectures on the low-resource Semitic language Amharic. My results demonstrate that while Transformer hybrids provide state-of-the-art language modeling accuracy, hierarchical recurrent models establish a new frontier of computational and memory efficiency.

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